



BIOLOGY CH.7 — CELL: BASIC UNIT OF LIFE — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway



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1. Cell Discovery & Cell Theory

- Cell ki khoj Robert Hooke ne 1665 mein ki thi - cork ke dead cells dekhe the microscope se. [Q418, Q456, Q463]
- Cell theory: Sabhi living things cells se bani hain. Viruses par ye applicable NAHI hai kyunki ye cells se nahi bane hote - Beijerinck ne 1898 mein viruses discover kiye. [Q418]
- Nucleus ki khoj Robert Brown ne 1831 mein ki thi. [Q460]
- Sabse chhoti unit of life jo independent exist kar sakti hai wo Cell hai - longest cell nerve cell, human mein largest cell female ovum hai. [Q456]
- Cytology (cell biology) Greek word 'kytos' (hollow container) se aaya hai. [Q463]

2. Prokaryotic vs Eukaryotic Cells

- Prokaryotic cells mein membrane-bound organelles NAHI hote hain - nuclear region ko nucleoid kehte hain, single chromosome hota hai, bahut chhote hote hain. Examples: Archaea, Bacteria, Cyanobacteria. Binary fission se reproduce karte hain. [Q431]
- Eukaryotic cells mein definite nucleus aur membrane-bound organelles hote hain - multicellular hote hain. [Q431]

- Prokaryotic ribosome: 70S (50S + 30S subunits). Eukaryotic ribosome: 80S (60S + 40S subunits). *[Q429]*
- Photosynthetic bacteria mein pigments plasma membrane ke infoldings mein hote hain - chloroplast NAHI hote. Plants mein Chlorophyll a, b aur Carotenoids chloroplast ke intrinsic part mein hote hain. *[Q453]*

3. Nucleus - Structure & Function

- Nucleus sabse bada organelle hai - cell ka 'command centre' aur 'master' hai, DNA store karta hai, double membrane bound hai. Nucleolus (kendrika) bhi isi mein hota hai. *[Q469, Q507]*
- Nucleus ke alawa Mitochondria aur Plastids mein bhi apna DNA aur ribosomes hote hain - ye semi-autonomous hain. *[Q436]*
- Nucleus 1831 mein Robert Brown ne discover kiya tha. *[Q460]*

4. Mitochondria - Powerhouse

- Mitochondria ko 'weird' isliye kaha jata hai kyunki inme apna DNA aur ribosomes hote hain - semi-autonomous organelle hai. *[Q420]*
- Mitochondria cellular respiration karta hai - pyruvate breakdown O_2 ke saath yahan hota hai, CO_2 + water + energy (ATP) banata hai. *[Q438, Q450, Q478, Q483]*
- Mitochondria double membrane bound, oval shaped hota hai. *[Q455]*
- Pyruvate (3-carbon) Krebs cycle mein 3 molecules CO_2 banata hai. *[Q438]*

5. Endoplasmic Reticulum (ER)

- SER (Smooth ER) - ribosomes nahi hote, lipids/steroid hormones/carbohydrates synthesize karta hai, liver cells mein drugs aur poisons ka detoxification karta hai. *[Q415]*
- RER (Rough ER) - ribosomes attached hote hain, proteins ka synthesis, folding, quality control aur despatch karta hai. *[Q415, Q475]*
- Smooth ER cell ke liye substances synthesize karta hai. *[Q442]*

6. Golgi Apparatus

- Golgi apparatus flattened stacked pouches (cisternae) se bana hai - proteins aur lipids ko transport, modify aur package karke vesicles mein bhejta hai. *[Q468]*
- Plant cells mein Golgi body ko dictyosomes kehte hain. *[Q486]*

7. Lysosomes - Suicide Bags

- Lysosomes ko 'suicide bags' isliye kaha jata hai kyunki ye biomolecules break down karte hain, waste remove karte hain, dead/damaged cells digest karte hain - sphere-shaped, eukaryotic animal cells mein paye jate hain. [Q437, Q455]

8. Ribosomes - Protein Factories

- Ribosomes ko cell ka 'protein factory' kaha jata hai - amino acids ko jodkar proteins synthesize karte hain (translation), RNA aur protein se bane hote hain, prokaryotic aur eukaryotic dono mein hote hain. [Q432, Q475]
- George Palade ne 1955 mein ribosomes discover kiye. [Q420]

9. Plastids

- Leucoplast - colorless/non-pigmented, roots mein paya jata hai, starch/lipids/proteins store karta hai. 3 types: Amyloplasts, Proteinoplasts, Elaioplasts. [Q424]
- Chloroplast - photosynthesis karta hai, green pigment chlorophyll contain karta hai. [Q448]
- Chromoplast - pigments contain karta hai. [Q424]

10. Vacuoles

- Plant cells mein vacuoles animal cells se badi hoti hain - ek single vacuole cell ke most interior space le sakti hai taaki water balance maintain ho. Animal cells mein vacuoles chhoti hoti hain aur waste sequester karti hain. [Q421, Q465]
- Vacuoles single membrane bound hote hain, tonoplast membrane hoti hai - water, sap, excretory products store karte hain. [Q421, Q442]
- Contractile vacuole Amoeba mein water regulate karta hai - osmotic pressure control karke water bahar pump karta hai. [Q425]
- Meristematic cells mein vacuoles ka absence hota hai - ye continuously divide karte hain. [Q454]

11. Cell Membrane / Plasma Membrane

- Plasma membrane proteins + lipids (phospholipids + cholesterol) + carbohydrates se bani hai - selectively permeable hai, dono plant aur animal cells mein hoti hai. [Q419, Q462]
- Selectively permeable membrane sirf certain molecules (specific solutes/solvents) ko allow karti hai - sabhi molecules freely diffuse nahi karte. [Q419]
- Gaseous exchange cell membrane ke across diffusion se hota hai - high concentration se low concentration ki taraf molecules ki movement. [Q444]

12. Cell Wall - Composition

- Plant cell wall: Cellulose + Pectin + Hemicellulose - polysaccharides se bani, water loss prevent karti hai, tensile strength deti hai. [Q445, Q467, Q486]
- Fungal cell wall: Chitin (glucans + glycoproteins + chitin) - ye components humans mein nahi hote isliye antifungal therapy ka excellent target hai. [Q487, Q500]
- Cork cells mein Suberin hota hai jo water/gas entry rokta hai - cork cells dead aur compactly arranged hote hain bina intercellular spaces ke. [Q477]
- Desert plants ki outer skin Cutin (waxy water-repellent substance) se covered hoti hai jo transpiration se water loss rokta hai. [Q479]
- Tough tissue Sclerenchyma dead (non-living) cells se bana hota hai jisme lignin-thickened walls hote hain. [Q488]
- Cell wall plant cells ko incoming water ke pressure ke against bursting se bachati hai. [Q443]

13. Cytoplasm & Cytosol

- Cytoplasm jelly-like fluid hai jo nucleus aur cell membrane ke beech hota hai - isme saare organelles suspended hote hain. [Q460]
- Cytosol cytoplasm ka aqueous component hai. [Q467]
- Cytoplasm metabolic processes host karta hai. [Q435]

14. Plant vs Animal Cell Differences

- Plant cells generally animal cells se badi hoti hain. Plant cells mein cell wall, chloroplast, large vacuoles hote hain - animal cells mein ye nahi hote. [Q430, Q465]
- Animal cells mein plastids NAHI hote. Plant cells mein centriole absent hota hai. [Q430, Q486]
- Dono mein common: cell membrane, mitochondria, ER, nucleus. [Q430]
- Human cheek cells (animal) vs Onion peel (plant) - main difference: plant cells mein cell wall hoti hai. [Q430]
- Root cells mein cell wall, cytoplasm, nucleus, vacuole dikhai dete hain - lekin chloroplast nahi kyunki root underground hai. [Q445]
- Leaves aur shoots mein large vacuole aur chloroplasts dikhai dete hain. [Q448]

15. Plant Tissues - Meristematic & Permanent

- Meristematic tissue: continuously dividing cells, growing regions (root tips, stem tips) mein payi jati hai. Term 'Meristem' Carl Wilhelm von Nägeli ne coin kiya. Types: Apical (length), Intercalary (elongation), Lateral (girth/thickness). [Q417, Q454, Q496]
- Meristematic cells mein vacuoles nahi hoti hain, thin cell walls aur dense cytoplasm hota hai. [Q454]

- Differentiation: meristematic cells se permanent tissue cells acquire karne ka process - permanent shape, size aur function. [Q498]
- Permanent tissues 3 types: Parenchyma (thin wall, food storage, living), Collenchyma (living, elongated, corners par thickened - cellulose+pectin, flexibility), Sclerenchyma (dead, lignin-thickened, hard/stiff). [Q499, Q503, Q506]
- Chlorenchyma: Parenchyma with chlorophyll (leaves mein, photosynthesis). Aerenchyma: Parenchyma with large air spaces (aquatic plants mein, buoyancy/floatation). [Q439, Q422]
- Aerenchyma Hyacinth (jal kumbhi) mein paya jata hai. [Q422]
- Plant tissue types: Dermal (outermost, protection), Vascular (xylem+phloem, transport), Ground (bulk, metabolic). [Q474]
- Collenchyma: living, elongated, irregularly thickened at corners - flexibility provide karta hai young plants mein. [Q499]

16. Animal Tissues

- 4 main animal tissues: Connective, Epithelial, Muscle, Nervous. Meristematic tissue sirf plants mein hota hai. [Q464]
- Epithelial tissue: body surfaces cover karta hai. Types: Simple (one layer), Stratified (more than one layer), Pseudostratified. [Q428, Q471]
- Simple epithelium exactly one cell thick hota hai. [Q471]
- Squamous epithelium: flat cells, capillaries/alveoli/glomeruli mein, rapid diffusion. [Q449, Q441]
- Cuboidal epithelium: equal width/height/depth, kidney ducts/pancreas/mammary glands mein. [Q449]
- Columnar epithelium: pharynx mein, goblet cells contain karta hai. [Q441]
- Ciliated columnar epithelium: trachea, bronchi, oviducts ko line karta hai. [Q441]
- Connective tissue: support/connect/separate karta hai. Examples: Blood, bone, cartilage, tendon, ligament, adipose tissue. [Q470, Q493, Q505]
- Blood connective tissue hai - bone marrow mein RBC, WBC, platelets bante hain. [Q470, Q472]
- Adipose tissue: fat store karta hai skin ke neeche aur internal organs ke beech, insulator ki tarah kaam karta hai. [Q440]
- Muscle tissue: 3 types - Cardiac, Skeletal, Smooth. Smooth muscle blood vessel ka diameter change karta hai. [Q508]
- Nervous tissue: brain, spinal cord, nerves mein - body activities coordinate karta hai. [Q440]
- Neuron (nerve cell): cell body (soma/cyton/perikaryon), dendrites (signals receive), axon (impulses transmit). Synapse: neuron ke beech connection. [Q447, Q489, Q491]
- Cyton mein Nissl's granules aur mitochondria hote hain. [Q489]
- Bone cells: Osteoblasts, Osteoclasts, Osteocytes, Osteoprogenitor cells. [Q447]

17. Cell Division - Mitosis

- Mitosis stages: Prophase (chromosomes condense, spindle forms), Metaphase (chromosomes equatorial plate par line up, spindle fibres attach), Anaphase (sister chromatids separate), Telophase (chromosomes uncoil, nuclear envelope reforms, cell divides). [Q458, Q502]
- Centromere: sister chromatids ko cell division ke dauran jodta hai. [Q501]
- Centrosome: sirf animal cells mein, 2 centrioles se bana, cell cycle regulate karta hai. [Q501]
- Nucleosome: eukaryotic chromosome ki structural unit - DNA histones ke around coiled hota hai. [Q501]

18. Cellular Respiration & Energy (ATP)

- ATP (Adenosine triphosphate) cell ki 'energy currency' hai - adenine + ribose sugar + 3 phosphate groups. [Q426, Q457, Q461, Q490]
- ATP mitochondria (powerhouse) mein synthesize hota hai - molecular transportation across cell membranes ke liye use hota hai. [Q490]
- Glycolysis: glucose $\rightarrow$ 2 pyruvate + ATP, cytoplasm mein hota hai. [Q423, Q435, Q446]
- Pyruvate breakdown using O_2 : mitochondria mein hota hai $\rightarrow CO_2$ + water + energy (Krebs cycle + electron transport chain). [Q450, Q478, Q483]
- Anaerobic respiration: cytoplasm mein hota hai (yeast/bacteria), incomplete oxidation se kam energy. [Q452]
- Aerobic respiration: mitochondria mein hota hai, complete oxidation se zyada energy. [Q452]

19. DNA, Genes, Chromosomes

- DNA 2 long polynucleotide strands ki double helix hoti hai - James Watson aur Francis Crick ko credit. [Q509]
- DNA genetic instructions carry karta hai proteins ki structure/function ke liye. [Q495]
- DNA transcribed hota hai RNA mein, jo protein synthesis ka template hota hai. [Q495]
- Genes: DNA ke segments jo heredity ki functional unit hain, parent se offspring mein transfer hote hain, body ki structure/function/metabolism control karte hain. [Q480, Q482, Q484, Q504]
- 'Genetics' term William Bateson ne 1905 mein coin kiya. 'Gene' term Wilhelm Johannes ne 1909 mein coin kiya. [Q504]
- Genotype = DNA mein genes ka set jo traits ke liye responsible. Phenotype = physical appearance. [Q484]
- Chromosomes: thread-like structures containing DNA wrapped around histone proteins. Humans: $2n=46$ (44 autosomes + 2 sex chromosomes). [Q482, Q509]

20. Protein Synthesis

- Transcription: mRNA DNA template se nucleus mein banta hai RNA polymerase se. [Q481]

- mRNA (messenger RNA) protein synthesis (translation) ka template hota hai. [Q481]
- Ribosomes protein factories hain jo amino acids jodte hain. [Q432]

21. Specialized Cells

- RBCs (Red Blood Cells): nucleus nahi hota, bone marrow mein bante hain, oxygen carry karte hain, CO₂ remove karte hain. Lifespan 120 days. Normal range: Male 4.7-6.1 million/mcL, Female 4.2-5.4 million/mcL. [Q492]
- WBCs: 12-20 days lifespan. Platelets: 8-10 days. [Q492]
- Neurons (nerve cells): branched, long, information receive aur transmit karte hain. [Q447]
- Rod aur Cone cells: retina (eye ke back) mein hote hain. Rod cells low light vision (rhodopsin pigment), Cone cells colour vision (iodopsin pigment). [Q459]
- Flame cells: Planaria (flatworm, phylum Platyhelminthes) mein excretory organs - solenocytes bhi kehte hain. [Q473]

22. Unicellular Organisms

- Paramecium: slipper-shaped, unicellular, pond water mein, cilia se covered. [Q433, Q451]
- Amoeba: entire cell surface se food leta hai pseudopodia se - single-celled, cell membrane, dense nucleus, vacuoles. [Q451]
- Paramecium specific spot (oral groove) se food leta hai. [Q451]
- Unicellular organisms: Bacteria, Protists, Yeast. [Q433]
- Bread mould (Rhizopus): multicellular fungus. Cuscuta: parasitic plant. [Q433]
- Muscles inherently multi-cellular hote hain — Unicellular (single-celled) organisms mein Bacteria, Protists, Yeast aate hain (prokaryotic-eukaryotic types mein categorize). [Q494]

23. Osmosis, Diffusion, Plasmolysis

- Osmosis: water ka movement semipermeable membrane ke across hypotonic se hypertonic solution ki taraf - hypotonic solution mein cells swell kar jati hain. [Q434]
- Plasmolysis: cell ka cell wall se contraction due to water loss (hypertonic solution mein) - jaise Rhoen leaf slide par strong sugar solution daalne se. [Q427, Q466]
- Deplasmolysis: jab water andar jata hai aur cytoplasm swell hota hai. Cytolysis: cell bursting due to osmotic imbalance. [Q466]

24. Other Cell Facts

- Ionizing radiation atoms ko affect karke DNA damage karti hai - direct aur indirect (water radiolysis) dono tarah se. [Q476]
- Benzene vapor anemia aur leukemia cause karta hai. Chloroform eyes/skin/liver/kidney harm karta hai. [Q476]
- Tissue: cells similar in origin, form and function. Histology = tissue ka study. Histopathology = disease-related tissue study. [Q497]
- Bone marrow mein stem cells se RBC, WBC, platelets develop hote hain. [Q472]
- Guard cells mein jab water jata hai → turgid (swollen) hoti hain → stomatal pore opens (CO₂ intake, O₂ release). [Q485]
- Leaf cross-section: Upper epidermis (water loss prevent), Lower epidermis (CO₂ in, O₂ out), Guard cells (stoma regulate), Air spaces (gas exchange), Chloroplast (light capture). [Q416]
- Plant waste storage: leaves (shed hote hain), old xylem (resins/gums), cellular vacuoles. [Q421]